

Degree Thesis

Computer Science & Engineering, 180 credits



Virtual Reality Data Visualization

Concepts, technologies and more

Degree thesis in Computer
Engineering, 15 credits

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Abstract

Data Visualization (DV) can be seen as an important tool for communication and data analysis. Especially when huge amounts of data are involved, visual representation of data can facilitate observation of trends and patterns as well as understanding. Currently, two dimensional displays are mainly used for Data Visualization, both in two and three dimensions (2D and 3D). However, two dimensional displays are limited in terms of 3D visualization because they do not allow for true sense of depth and do not cover the observer's full Field Of View (FOV). An alternative approach is to use Virtual Reality (VR), which provides an immersive and interactive 3D environment. VR has been mainly used for gaming and simulated training. However, other areas are now emerging because VR technologies became relatively affordable. For example, one possibility is to explore VR for DV and this was the main goal of this project. To accomplish that, a literature study was performed to identify terminologies and definitions, hardware and software technologies, techniques and examples in the fields of DV and VR. In addition, in order to exemplify DV through VR, a prototype system was implemented using Unity 3D, a leading engine for VR. To visualize the developed VR environment, a HTC Vive Head Mounted Display (HMD) was used. The developed prototype system can display data from a local dataset in a scatter plot with three axis in VR. In the virtual environment created by the system, the user can select the attributes in the dataset to be displayed by the 3D scatter plot. Once the data is plotted, the user can use the handheld joystick to move, rotate, tilt and scale the scatter plot. Achieved results indicate immersion and interaction as the main perceived benefits of DV using VR.

Sammanfattning

Datavisualisering (DV) kan ses som ett viktigt verktyg för kommunikation och dataanalys, speciellt när stora mängder data behandlas. Visuellt representation kan främja observationen av trender och mönster samt förståelsen av datan. För närvarande används tvådimensionella displayer huvudsakligen för datavisualisering, både i två och tre dimensioner (2D och 3D). Emellertid är tvådimensionella displayer begränsade i 3D-visualisering eftersom de inte möjliggör äkta djupseende, och täcker inte observatörens fulla synfält (Field Of View (FOV)). Ett alternativ tillvägagångssätt är att använda Virtual Reality (VR), vilket tillhandahåller en omslutande och interaktiv 3D-miljö. VR har huvudsakligen används för spel och simulerad träning. Däremot börjar nya användningsområden uppstå då VR teknologin har blivit mer prisvärd. Ett användningsområde är VR för DV, vilket var det huvudsakliga syftet för det här arbetet. För att uppnå syftet utfördes en litteraturstudie för att identifiera teknologier och definitioner, hårdvaru- och mjukvaruteknologier, tekniker och exempel inom området av DV och VR. Dessutom, för att exemplifiera DV genom VR, utvecklades ett prototypsystem. Vilket implementerades genom Unity 3D, en av de ledande spelmotorerna. För att visualisera den utvecklade VR-miljön användes ett HTC Vive Head Mounted Display (HMD). Den utvecklade prototypen kan visualisera data från ett lokalt dataset genom ett spridningsdiagram med 3 axlar, i VR. I den virtuella miljön som skapas av systemet tillåts användaren att välja attribut från datasetet för att sedan visualisera dessa genom det tredimensionella spridningsdiagrammet. När datan väl är visualiserad, kan användaren använda de handhållna kontrollerna för att flytta, rotera, luta och skala grafen. Uppnådda resultat indikerar på omslutning och interaktion som de huvudsakliga fördelarna av DV genom VR.

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Acronyms & Glossary

CPU Central Processing Unit.

DoF Degrees of Freedom.

DV Data Visualization.

et al. abbreviation for et alia, meaning: and others. Is used in formal writing to avoid a long list of names of people who have written something together.

FOV Field Of View.

FPS Frames Per Second.

GPU Graphics Processing Unit.

HMD Head Mounted Display.

IMU Inertial Measurement Unit.

VCS Version Control System.

VR Virtual Reality.

WMR Windows Mixed Reality.

1 Introduction

Each year the computers becomes increasingly complex and they collect data in vastly amounts. There is a demand to visualize data to get a better understanding of what's happening. To extract and understand relevant data from a huge collection of data is a problem for many companies. Visualized data shows data in a compact and coherent manner. Only parts of the data can be visualized in a 2D graph due to restrictions with two dimensions, where the big picture and the connections between different graphs can be hard to see. There are ways to visualize data in a 3D graph on a 2D screen to get a better understanding of the data. Unfortunately there are limitations of displaying data this way, such as lack of depth in a 2D screen. 3D models often have few ways of interacting in a easy manner. VR will incorporate the 2D and 3D models in its simulated environment to enhance the interacting and viewing from a 2D screen.

The idea for this project was proposed by Capgemini [1], a company in the management and IT domain which provides consulting, technology and outsourcing services. The company would like to provide a solution that enables Data Visualization (DV) using Virtual Reality (VR). The main motivation for the company is that such an approach could enable a new service for their customers. However, Capgemini is new to the field of DV-VR and would like to get insights about the field.

1.1 Purpose

The purpose of this project is twofold. The first purpose is to provide Capgemini with an overview about the different technologies and possibilities of DV through VR. To reach this purpose, an identification about existing techniques, frameworks, hardware and software technologies are going to be performed. The second purpose of this project is to develop a prototype system for Data Visualization through Virtual Reality, to present to Capgemini as an example of what this can look like. The prototype is going to be developed using the knowledge that has been gained during the reach of the first purpose. However, the company has no specific dataset to visualize or any restrains on how the data should be visualized.

1.2 Problem formulation

Problem formulation regarding the project are as follows:

- What is Data Visualization?
- What is Virtual Reality?
- How can DV be done through VR?
- Which products are available on the market today?
- What platforms are there for VR-development?
- Is interaction needed? And if so, what kind?
- How to help unfamiliar users within the system?

1.3 Delimitation

This project includes a prototype system of DV through VR. However this project will not include: Modification of the data itself when the graph is displayed, or multi-user viewing with several headsets.

1.4 Related work

There are mathematical tools, such as Mathematica and MATLAB that can display data in 1D length, 2D area and 3D volume [2, 3]. There are also Data Visualization software and analytics tools like Tableau [4] and Qlik [5]. But their limitation is that the data is projected on a 2D screen with few methods of manipulating the graph.

A study by Donalek et al. at Caltech and JPL [6] shows proofs of concept of DV in VR, as seen in figure 12. Their study also showed that when users was immersed in their data they were prone to do less errors, and that the discovery process is much more powerful when they can interact with each other. Many of them have now moved on and created Virtualitics Immersive Platform, figure 1.4 [7] - a collaborative data exploration platform that merges Artificial Intelligence, Big Data and Mixed Reality.



Figure 1: A collaborative VR environment [7].

Paul Sullivan [8] master thesis user study testing effectiveness of his developed VR-graph, figure 2, compared to 2D line charts on paper. Unfortunately the results shows that his participants where both quicker and more accurate using the traditional type of Data Visualization (DV). Although most of the test subjects where new to VR and not familiar with either the environment or the type of graph. Moreover a keyboard was used for moving around in the environment, controllers that is visible with tooltips could have helped. Lastly almost all of the was users excited about the method of visualization in VR.

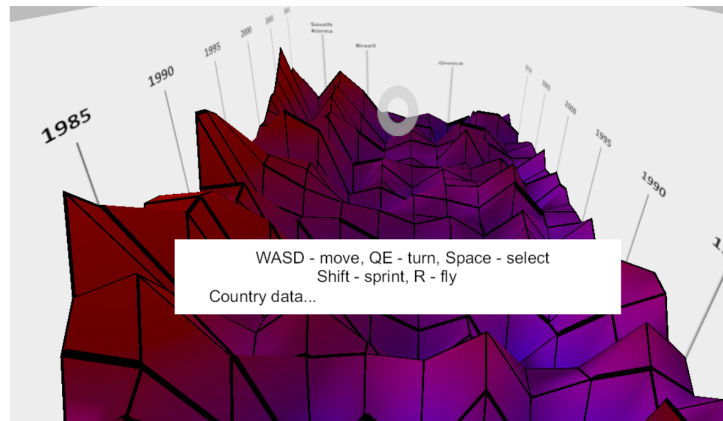
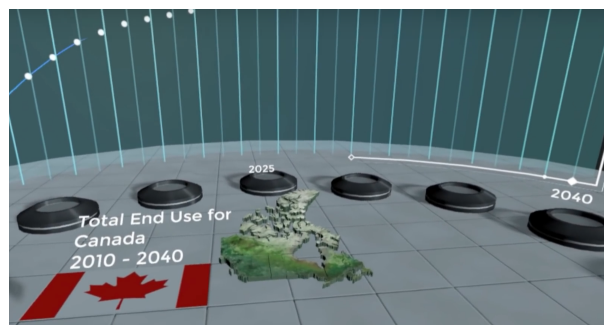
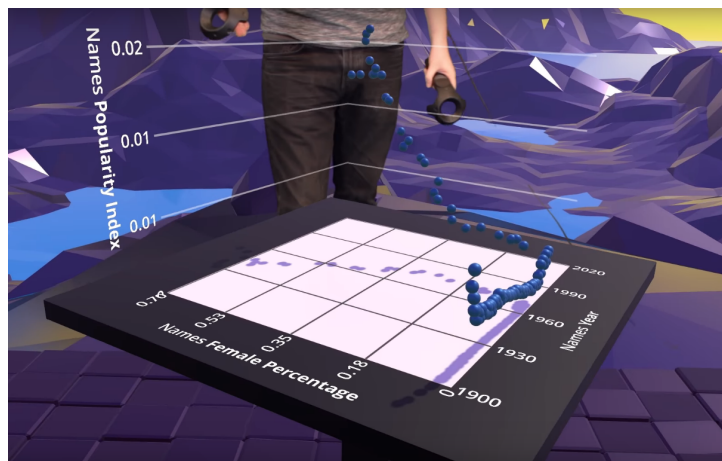


Figure 2: A picture of Paul Sullivans VR-system [8].

Other similar solution also exist from Nirvaniq Labs [9] and Looker [10]. In figure 3(a) Canada's energy demand from 2010 to 2040 are shown in VR but is utilized in different standard 2D graphs. In figure 3(b) a scatter plot is seen, plotting the name Riley's popularity over a period of time. The visualization is made by the company Looker.



(a) Nirvaniq Labs displaying Canadas Energy use in an interactive way [9].



(b) A scatter plot from Looker [10].

Figure 3: Data Visualization in VR from two leading companies in the field.

To conclude the related work, the investigated reports have been using or are recommending an supportive game engine like Unity 3D instead of OpenGL. Since this will speed up the development process and increase the support for hardware components like controllers and such. Only Paul Sullivan has used a seated VR-environment and a PC-keyboard to orient inside the environment. The others have used controllers that are visible inside the environment. Different types of graphs have been used for visualization.

1.5 Outline

The remaining part of this report is organized as follows. Chapter 2 theoretical background, explains the different concepts related to this work, such as DV, 2D-visualization, 3D-visualization and VR. Moreover different frameworks/platforms and techniques for VR is explained. Chapter 3 method, explains development method and why different decisions was made. It also describes different functionalities within the prototype together with software and hardware specifications. Chapter 4 contains the result, here the result of the prototype is shown and explained. Chapter 5 contains the discussion part, here the students discuss the projects implementation and result. In Chapter 7 a conclusion about the project is written followed by future optimization. At the end of the report bibliography and appendix are added.

2 Theoretical background

In this chapter Data Visualization, 2D and 3D visualization and Virtual Reality are explained. The visualization techniques are followed by different technologies and game engines for VR.

2.1 Data Visualization

Data Visualization (DV) is the graphical display of abstract information. It has two purposes, communication and sense-making(data analysis). Important information is often stored in data, and data visualization is a powerful way of discovering and understanding of that information. Some of that information is abstract, since it describes things that are not physical, like statistical information. Even though it doesn't belong to the physical world, it's still possible to display it visually.

To do this, we must find a way to give form to things that have none. The translation of abstract into visible physical attributes (such as length, size, color, position and shape) can only thrive if we understand a bit about cognition and visual perception. In other words, to visualize data in a effective way, one must follow design principles that derive from a understanding of human perception [11].

2.1.1 Data Visualization and Human Perception

The effectiveness of data visualization comes from its way of shifting the balance between cognition and perception, thereby taking fuller advantage of the brain's abilities. Cognition (i.e. thinking) is handled by the cerebral cortex, located in the front of the brain is quite slow and not so efficient. Visual perception (i.e. seeing) is handled by the visual cortex, which is located in the rear of the brain, is much faster and more efficient. Traditional data presentation and sense-making methods require conscious thinking for most of the work figure 4(a). DV shifts the balance better, using a greater part of visual perception figure 4(b)

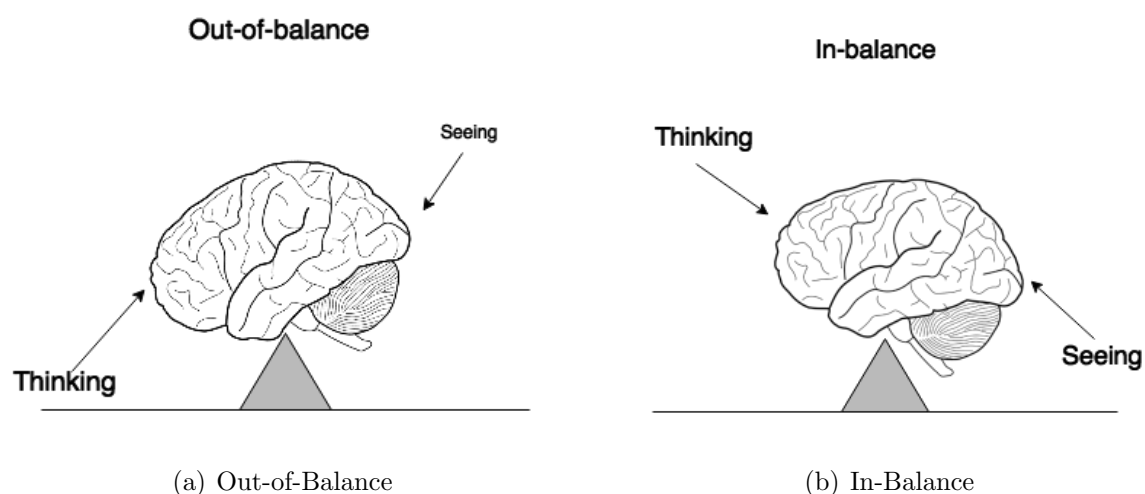


Figure 4: Traditional data presentation and sense-making methods require more thinking and less seeing (a), while Data Visualization shifts the balance better (b). The picture is a recreation of [11] made by the students.

One of the greatest strengths of DV is our ability to process visual information much quicker than verbal information. Good DV techniques and technologies used in a proper way, can extend our thinking into new realms of analytics [11].

2.1.2 2D Visualization

Here are some strength and weaknesses with 2D Visualization, accordingly to S. Dübel et al. [12] and one of the most well-known chart of all time, figure 7, [13].

Basic Principles - Data is presented on a 2D reference space with 2D objects as seen in figure 5. 2D visualization is widely used and are well known.

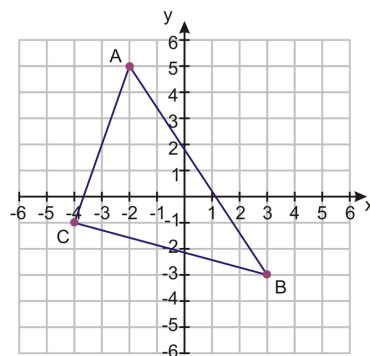


Figure 5: Basic principle of an 2D graph [14].

Strengths - Techniques like maps on a smaller scale, histograms, pie charts and different kinds of graphs are excellent display in 2D. Using a monochromatic color scheme, makes it much easier for the eye to understand. As figure 6(a) shows. It is general more suitable for precise measurement and interpretation.

Weaknesses - Maps on a larger scale gets distorted, because the curvature of the earth. For example the world map in every classroom in the world is distorted, because displaying spherical objects flat surface expands and contracts different regions of the map, see figure 6(b).

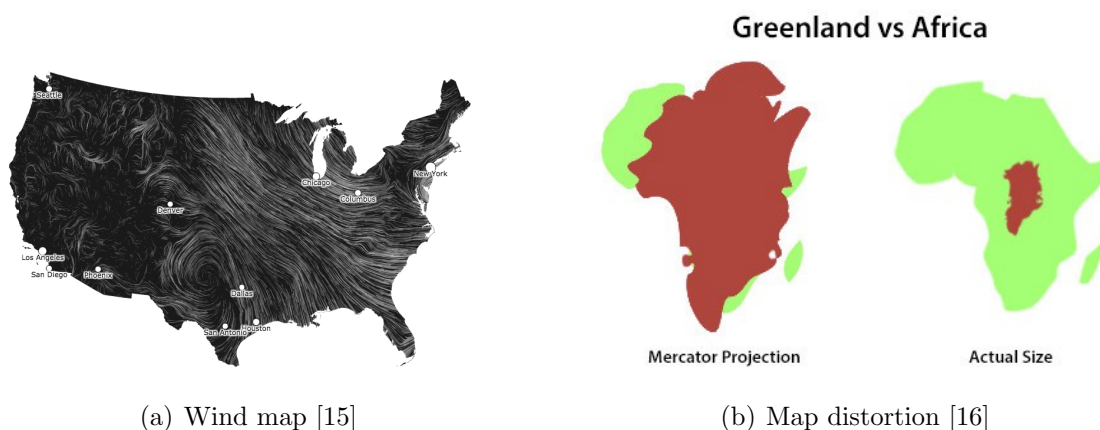


Figure 6: (a) shows a wind map , while (b) shows how a map gets distorted.

Situations where most appropriate - Displaying simple and basic information, but with the correct structure very powerful graph and diagrams can be created. One of the most well-known graphic is the illustration of Napoleon's army marching from Polish/Russian border to Moscow and back, 1812-1813. This is displayed in figure 7. It contains six types of data in two dimensions. It illustrates Napoleon's troops, distance, temperature, longitude and latitude, direction and location at specific dates.

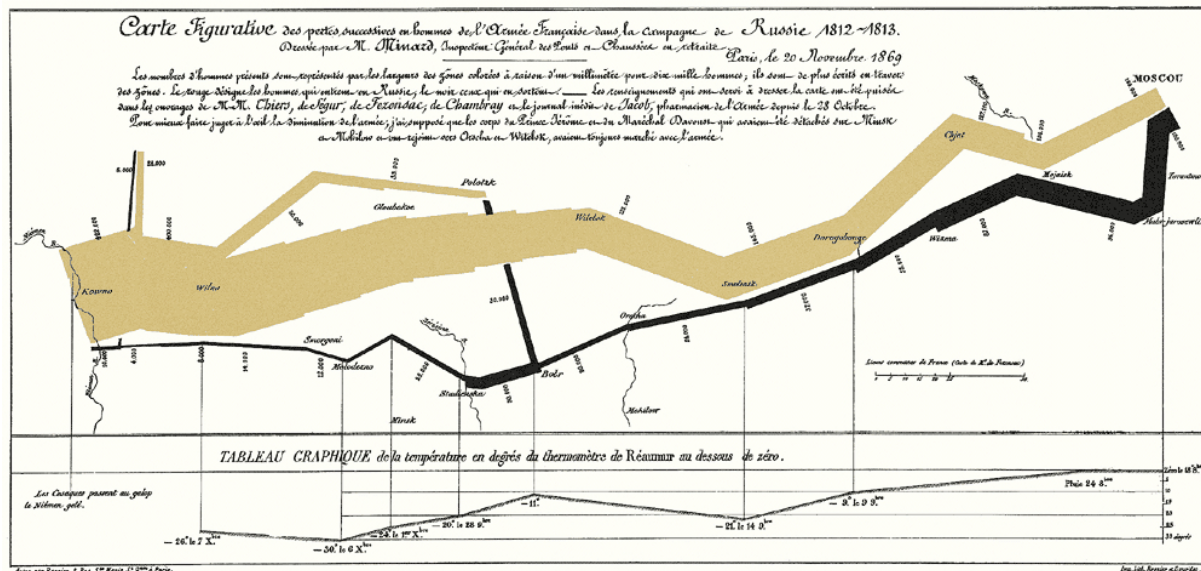


Figure 7: Excellent data visualization of Napoleon’s Russian campaign 1812 using 2D [17].

Situations where least appropriate - When presenting complex data and several different parameters it may require a 3D graph to fully grasp the complexity of the object being presented.

2.1.3 3D visualization

Drew Skau, a PhD CS Visualization student at UNCC with an undergrad in Architecture, states the following weaknesses data visualization in 3D [18] as explained in Weaknesses.

Basic Principles - Data is presented with 3D objects in a 3D reference space as seen on in figure 8. The 3D object is projected to a 2D screen.

Strengths - Can display more and complex data, such as node trees and scatter plots.

Weaknesses - Displaying data with a diagram or chart that doesn't have a third dimensional component. Pie charts and maps are two example that does not represent well in 3D. Pie charts uses angles and area to display data, when displaying a pie chart in 3D the angels and area get distorted. As we can see in figure 9(a) comparing a pie chart in 2D and 3D, there are a clear distortion between the two.

Figure 9(b) is a map where the height of the regions is connected to data, it can have some problems with some regions obscuring other regions. Extracting data from regions on the map is difficult since it has no proper scale, and compare two different regions is

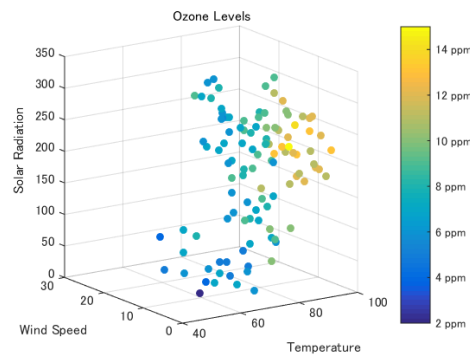


Figure 8: Basic principle of an 3D graph [19].

also difficult. For 3D visualization to be valuable it must be viewed from multiple angles or have to be interactive so that they can be rotated at will. When the data density is high, some occlusion may occur in the 3D presentation. But with correct filtering the occlusion can be reduced.

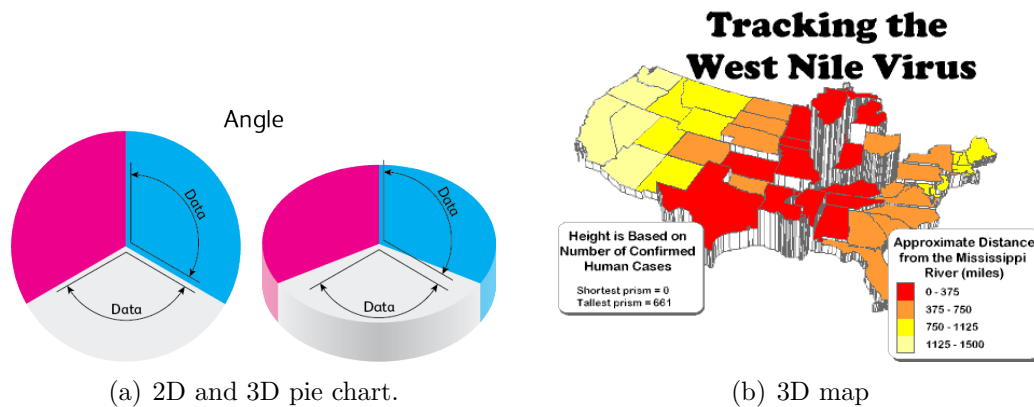


Figure 9: Bad data visualization example using 3D [18].

Situations where most appropriate - When analyzing complex graph structures, figure 10(a), and when performing strength analysis as figure 10(b) shows.

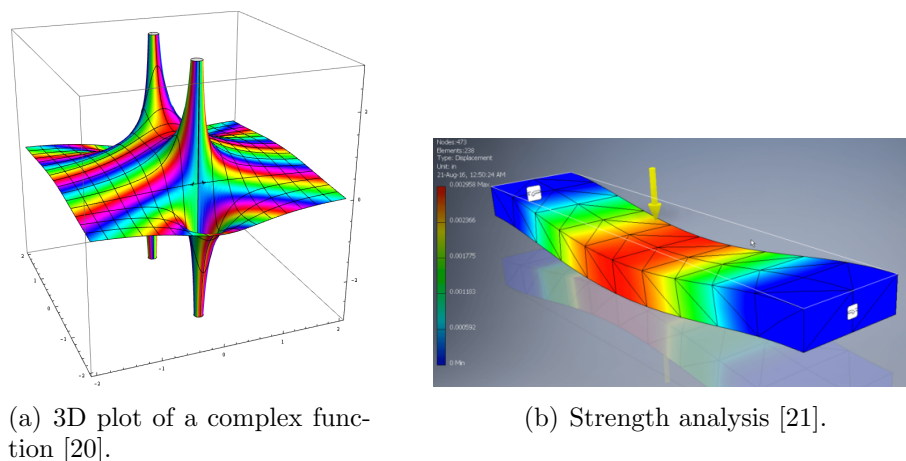


Figure 10: Good data visualization example using 3D.

Situations where least appropriate - Displaying text and histograms with high variance, causing some bars to obscure others.

2.2 Virtual Reality

In this subsection VRs basic principles, strengths and weaknesses are explained, followed by some examples. Furthermore HMDs are explained, as is, immersion, interaction and cybersickness in VR.

2.2.1 Basic Principles

There are some different definitions of VR but they are essentially equivalent. They all mean that VR is an interactive and immersive (with the feeling of presence) experience in a simulated (autonomous) world [22] as seen in Zeltzer's cube, figure 11, which can be used to determine the level of advance of VR systems. The virtual world can be solely made up (cyberspace), a simulation of a real but remote environment (telepresence) or a combination of the two [23].

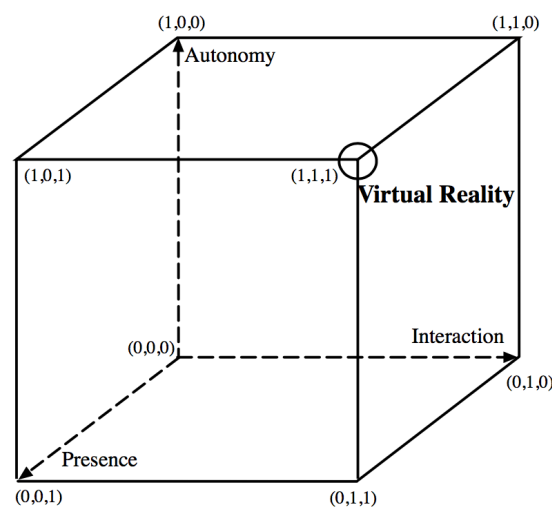


Figure 11: Zeltzer's cube (adapted from [22]) - Autonomy, interaction and presence in VR.

2.2.2 Strengths

Some of VRs strengths are: Immersive, interactive and collaborative multi-user visual data exploration, figure 12. In addition the technology is rapidly getting improved each day by video gaming and other entertainment industries. Moreover it's more portable and cheaper than the previously multi-million dollar CAVE-type (an immersive VR environment where the environment are projected to between three and six of the walls of a room-sized cube.) installations with the same capability to explore DV [6]. Furthermore Ware et al. indicates that displaying data in 3 dimensions instead of 2 can make it easier to understand, and that up to 3 times as much information can be perceived with a HMD compared to a 2D view [24].

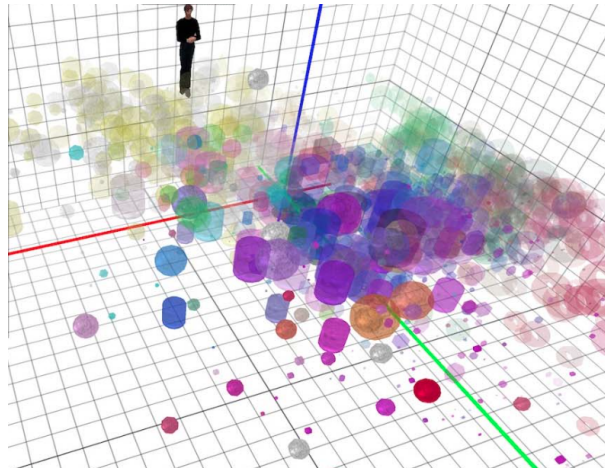


Figure 12: Immersive and collaborative DV-environment created at Caltech, where different data parameter values are mapped to the displayed XYZ, data points shape, colors, sizes and transparencies in a scatter plot [7].

2.2.3 Weaknesses

Some of VRs weaknesses are: The presentation of textual information, such as equations, formulas, or definitions. The problem is compounded by the fact that users may choose any viewpoint they wish [25]. It requires a computational powerful computer and a HMD which can be very expensive. Furthermore the using of powerful HMDs restricts the users movements, since it's wired to the associated computer, which also might expose the user for hazards outside the environment. Furthermore the HMD also might cause cybersickness to the user. Cybersickness is a form of motion sickness with symptoms reported to include nausea, vomiting, eyestrain, disorientation, ataxia, and vertigo [26, 27, 28].

2.2.4 Situations where most appropriate

VR is suited for several types of DV, including: immersive and collaborative DV of remote places, when interacting with the data and when studying multi-dimensional complex datasets. Moreover it suits full-motion interactive gaming very well [6, 27]. Furthermore it's a good platform for immersive simulators for education and training [29, 30, 25]. It's also suited for studying multi-dimensional complex datasets and for Big Data visual analytics [27, 31, 6].

2.2.5 Situations where least appropriate

VR is not so well suited for displaying textual information, such as equations, formulas, or definitions [25].

2.2.6 Examples

Here follows some examples on how VR can be used. Giving new capabilities to Tele-robotic operators, planetary scientists and Archaeologists exploring the surface of Mars. Giving them the opportunity to interact both with the data and with their colleagues [6]. Training excavator personnel using simulator with joysticks, platform and HMD(VR-excavator) [29]. Furthermore Wijewickrema et al. [30] shows how advanced surgical education, in their case temporal bone procedures, such as cochlear implant surgery can be performed in VR. Their study explains several advantages of doing surgical training and pre-planning in VR compared to traditional methods. Traditional training relies on cadaveric dissection and supervised training in the operating theatre. However, both these forms of training have become inefficient since it takes up surgeon's time and the scarcity of cadavers. The VR environment offers repeated practice in a risk-free environment, along with automated guidance and assessment. It can also be used for immersive visualization for Materials Science Data Analysis [27] and for realistic virtual models for factory layout planning [32].

Furthermore Vincur et al. [33] presents a software visualization tool that utilizes the modified city metaphor to represent software system and related analysis data in a VR environment (figure 14). The city metaphor in short are: Each street represents a certain package. Contained packages form branching streets. The system level is represented by a main road from which the top-level packages branch off. Classes are represented as square building plots. Vincur et al. has modified deeper steps of the metaphor: "In resulted city, classes are represented by buildings and methods are represented by floors. Each floor consists of number of blocks (cubes) equal to value of chosen software metric", "one can easily distinguish between buildings that represent interfaces (skinny and tall), abstract classes (skinny at the top) and ordinary classes" Gadepally et al.[31] (figure 13) shows immersive and interactive urban planning combining NYC taxi transportation data and twitter feeds. Lastly Wang et al. [34] brings movements and health into their VR-system by providing VR-Rides. VR-Rides consists of 3 different types of rides, where each one of them are built on the same ground, having a HMD for visual immersion, showing images from Google street view, measuring the user through some sort of sensor or physical activity device and providing movement input from the user through a recumbent trike or DeskCycle.

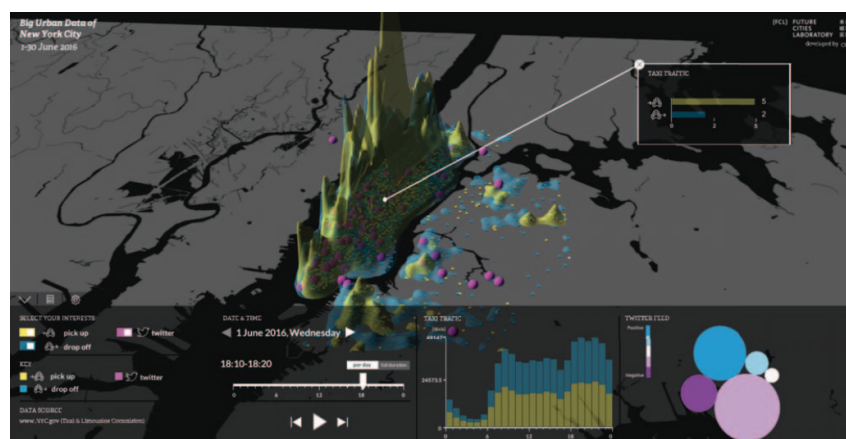


Figure 13: Transportation data and social feeds of New York City(NYC) combined to help urban planners in new immersive ways of understanding[31].

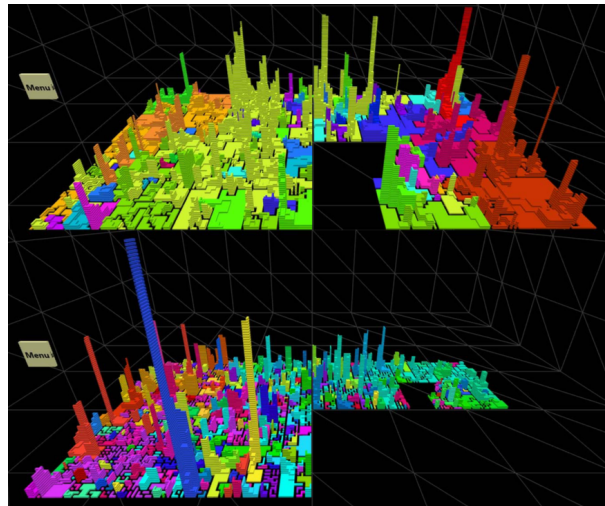


Figure 14: VRCity used for software analysis in VR, showing analysis of JUnit(top) and JHotDraw(bottom)[33].

2.3 VR Technologies

In this subsection the different types of HMDs and how they work are explained. Furthermore the different levels of immersion and interaction are described.

2.3.1 VR hardware

Head Mounted Display (HMD) is a stereoscopic display that's used to display one image in front of each eye creating a sense of depth, allowing the user to be immersed in the 3D environment created [35]. Whereas the first HMD states back to 1968, modern HMDs provides state of the art graphics and sensors. Depending on the type of HMD interaction can happen through head tracking, eye tracking or motion tracking, allowing the user to move or interact in the virtual world in real time.

Smartphone HMDs - Smartphone HMDs are a headset used to hold a smartphone in front of the users FOV to display information. There are simple and cheap ones as the *Google cardboard* as seen in figure 15(a) and more robust and comfortable ones like *Google Daydream View* displayed in figure 15(b) or *Samsung Gear VR* in figure 15(c). Some of these HMDs comes equipped with a controller to extend interaction. For these HMDs the smartphones own Inertial Measurement Unit (IMU) consisting of accelerometer, gyroscope and sometimes magnetometers is used for rotational tracking, except for Samsung Gear VR, which has its own internal IMU.



Figure 15: Smartphone HMDs from simpler (a) to more robust ones (b) and (c).

PC HMDs - Together with the IMUs that's used for rotational tracking, PC driven HMDs also uses Positional tracking [39] for orientation in the room. There are typically 2 types of positional tracking, Outside-in as seen in figure 16(a) and Inside-out figure 16(b). Platforms such as *Oculus Rift* and *HTC Vive* as seen in figure 17(a), are based on outside-in tracking, where external beacons or trackers are needed. Vive uses SteamVR Lighthouses which act as beacons to communicate with the infrared receivers on the HMD and accessories such as the handheld controllers. Rift utilizes external optical sensors called Constellation that detect the IR LED markers on the HMD and its accessories. Inside-out tracking typically uses time of flight sensors or stereo vision to detect and map the change in relative position. Marker-less inside-out tracking uses simultaneous localization and mapping (SLAM) algorithm to constructs or updates a map of an unknown environment, while simultaneously keeping track of the user's location. Which is used in Windows Mixed Reality (WMR) HMDs like *Samsung Odyssey* in figure 17(c) and *HP* in figure 17(b).

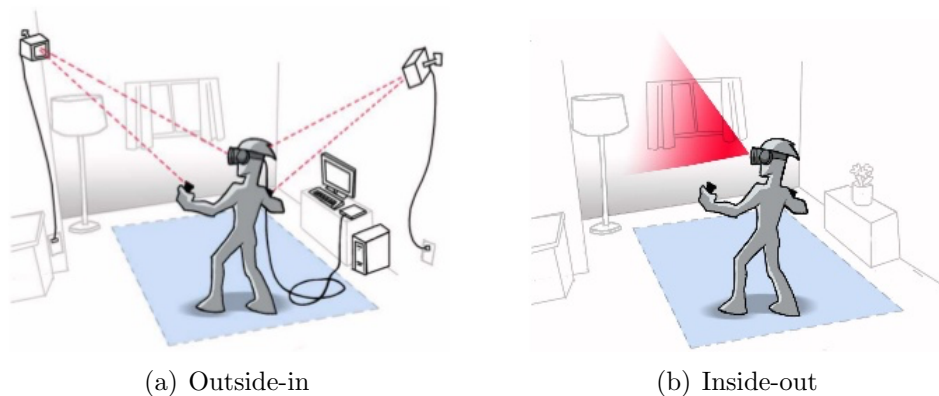


Figure 16: The 2 types of positional tracking, Outside-in (a) and Inside-out (b) [40].

The advantage of outside-in tracking is that the beacons or markers provide explicit signaling/patterns, which simplifies the tracking computation. There is less data to be processed, which lowers the computing power required and the tracking accuracy tends higher. On the down side, the user must remain in the tracker/beacon's line of sight to avoid occlusion. Inside-out tracking has simpler setup but it requires more complex vision processing and computing performance. All PC driven HMDs are today wired to a PC, which restricts the users mobility. Wireless transmission between the PC and the HMD is under development.



Figure 17: 3 different PC driven HMDs from different manufactures (a) HTC, (b) HP and (c) Samsung.

Standalone HMDs - During 2018 several standalone HMDs are being released, *Oculus Go* and *Lenovo Mirage Solo* are 2 of them. These are a hybrid between smartphone HMDs and PC HMDs. Having its own CPU and LCD displays, making it as wireless as smartphone HMDs. With inside-out tracking and 6 DoF. Yet it will not have the same computational power as a standalone PC. The frequency of *Lenovo Mirage Solo* will be 75Hz and the frequency of the *Oculus Go* will be 72Hz. It is slightly lower than the 90Hz needed to not induce any cybersickness. Oculus will additionally have Fixed Foveated Rendering. Full Foveated Rendering detects where the eyes are looking and instructs the application to only render the center of eyes view in full detail. With less details rendered the further out, which will save significant clock cycles. Since Oculus doesn't support eye tracking, an assumed/fixed central point of gaze will be used [44, 45].

2.3.2 VR capabilities

The Degrees of Freedom (DoF) of an object represent its ability to move around in space. In a 3D space there are 6 DoF. Three represent rotational movement, pitch, roll and yaw. The other three represent translational movement, surge, heave and sway, as seen in figure 18 [46]. In VR 3 DoF usually means the 3 rotational movements, moving along the x, y and z axis. Smartphone driven HMDs are limited to these 3 DoF, allowing the user to look in every direction around them. But with no capability to move in the scene by moving in real life. PC driven HMDs are capable of 6 DoF. Allowing the user not only to look around, but also to move around in the virtual world.

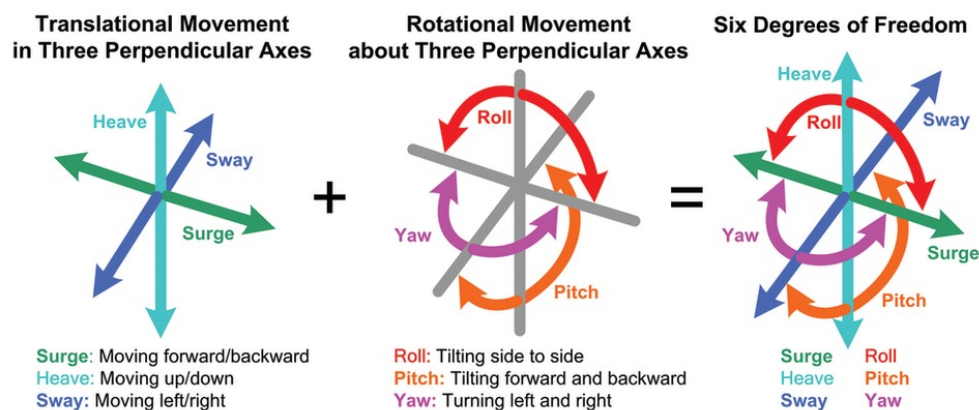


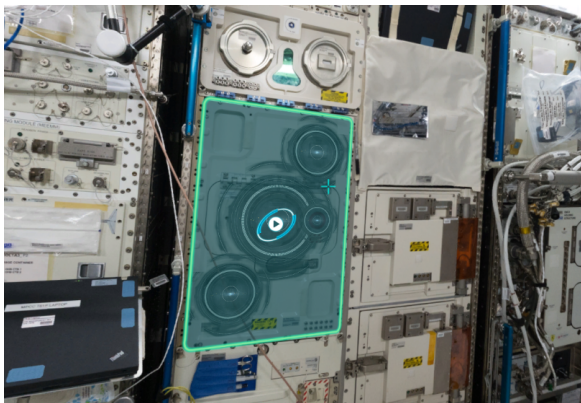
Figure 18: Combined translational and rotational movement showing the 6 Degrees of Freedom (DoF) from [46].

Immersion and interaction In a virtual environment system a computer generates sensory impressions that are delivered to the human senses. The type and the quality of these impressions determine the level of immersion and the feeling of presence in VR. These systems can be grouped in 3 levels according to the level of immersion that they offer [23]:

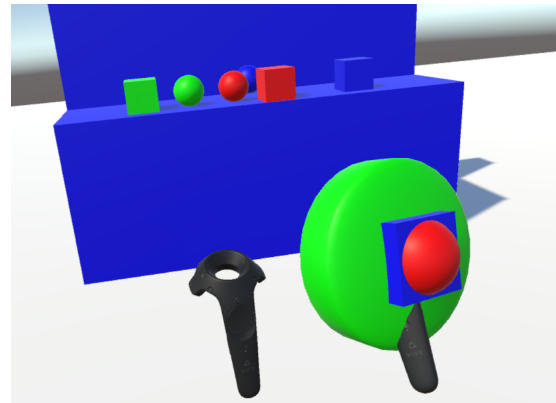
- **Desktop VR** - is the simplest type of VR. It uses a conventional monitor to display the image (generally monoscopic) of the world. No other sensory output is supported.

- **Fish Tank VR** - supports head tracking and thereby immersion. Still uses a conventional monitor (very often with LCD shutter glasses for stereoscopic viewing) but generally don't support sensory output.
- **Immersive systems** - the ultimate and nowadays generally used version of VR systems. They let the user totally immerse in computer generated world with the help of HMD that supports a stereoscopic view of the scene accordingly to the user's position and orientation.

Interaction in VR can be split up in 3 different ways: *Passive* - users sees, hears and feels the environment. *Exploratory* - user can explore the environment him/her-self. *Interactive* - user can explore, interact, and modify the environment. Interaction is usually supported by the HMD and can be further supported with some sort of controller. Figure 19 shows two examples of an *Interactive* VR-environment. In (a) a crosshair is in the center of the users viewpoint, here interaction happens if it's pointed at specific things for a specific time (see progress circle in the middle of the green rectangle). In (b) grabbing interaction is done with help of a controller and the VRTK asset.



(a) International Space Station tour in VR where interaction is done by the crosshair in the users focus view.



(b) Grabbing an object in VR using a HTC Vive controller.

Figure 19: (a) shows interaction by head tracking for a specific time, (b) shows interaction with a controller by VRTK, a Unity 3D asset, both pictures are taken by the students.

2.4 Cybersickness

Micheal Abrash [26] at Valve has outlined hardware factors that must be addressed to avoid VR-sickness. He describes the need for each of the following: Wide field of view, adequate resolution, low pixel persistence, high refresh rate of at least 90 Frames Per Second (FPS), displays illuminating all pixels simultaneously (global display), excellent optics and capabilities for optical calibration, precise tracking, minimal latency. Moreover Drouhard et al. [27] noticed that latency has a significant effect on user experience. "When rendering lags, user discomfort rapidly increases, which degrades performance and, ultimately adoption potential. In the context of big data, the latency effect will be a key challenge to overcome. The most viable strategy to overcome latency is through optimized rendering algorithms and multi-scale representations running on powerful graphics hardware."

2.5 Development platforms for VR

Here is an explanation of the different game engines and techniques, their language support and accessibility. Each of the three game engines support plugins through their respective website. The assets range from a wide variety, covering everything from textures, models and animations to whole project examples and tutorials. Unity have the most extensive asset store among the other game engines.

2.5.1 Unity 3D

Unity 3D is developed by Unity Technologies, a Danish company that later moved to San Francisco. The initial release was in 2005 exclusively for macOS. The game engine supports 2D and 3D graphics on 28 platforms including VR [47]. Unity supports two languages, JavaScript and C#, although JavaScript is deprecated as of August of 2017 [48]. All Unity plans are royalty-free and include all platforms, core engine features, continuous updates, and beta access. Unity offer a free version, called Personal, for beginners, students and hobbyist that include all core features.

2.5.2 Unreal Engine

Unreal Engine was initial released in 1998 and developed by a American company Epic Games. The game engine was developed for first-person shooters but has later evolved to support many genres. Unreal Engine supports projects to be deployed on 17 different platforms [49]. UnrealScript was the scripting language for the early versions of the engine, it was an object-oriented language that got deprecated in 2012 with the release of Unreal Engine 4, which is the current version. Now it only support C++ and a visual scripting called Blueprints Visual Scripting. The Unreal Engine is free to use but uses a 5% royalty fee on gross product revenue. The full C++ source code for the engine is posted on GitHub and available for those who have an account and installed the engine.

2.5.3 CryEngine

CryEngine is developed by the German company called Crytek. CryEngine 1 was ground-breaking when it was released in 2002 with its Sandbox editor, real-time lighting, long view distances and realistic vegetations. CryEngine V which is the current version added support for Virtual Reality. All of CryEngine, including the full engine source code, is available under a 5% royalty based model. The game engine supports development in either C# or C++.

2.5.4 OpenGL

Open Graphics Library, OpenGL, is not a game engine as the previous are, it is a cross-platform standard and API for defining 2D and 3D graphic images. OpenGL contains over 250 specific set of commands. The purpose of OpenGL is to communicate with the graphics card trough the commands to draw complex 3D scenes. The API is used to hide complexities of interfacing with different 3D accelerators by presenting a single and uniform interface. Each graphics card differ from each other between the different manufacturers, in a sense they speak different languages, by utilizing OpenGL programs can be written to serve all graphics card. OpenGL is managed by the non-profit technology consortium Khronos Group.

2.6 Theory summary

Chapter 2 explains how Data Visualization works and how users can benefit from it. Furthermore it defines what 2D, 3D and VR visualization is and the various advantages and disadvantages with them. Moreover in section 2.3, information about different types of HMD, tracking technologies and various levels of immersion and interactions. In section 2.5 the different development platforms are summarized and explained.

Regarding the information extracted in the theory, it is most beneficial to use an game engine for development. Unity 3D is the biggest among them and have a large communities that can help when an issue occurs. A PC HMD is also recommended to get maximum of Degrees of Freedom, immersion, interaction and processing power. VR has many strengths but also some drawbacks. The biggest benefits of using VR is the immersion and the true sense of depth. The weakness of using VR is that 2D objects such as text is represented poorly, due to the fact that they are 2D objects. This can be worked around if the 2D objects always are facing the HMD, but some occlusion can occur. Visualizing 3D object in VR may not always be beneficial either due to some distortion or occlusion can occur when choosing either an pie chart or an histogram. Moreover can VR cause cybersickness like nausea and disorientation.

3 Method

This chapter describes how the prototype was developed and what hardware and software was used. Furthermore it covers the functionalities of the prototype.

3.1 Literature study

To get a better understanding about DV, VR and similar solutions and to be able to present these to Capgemini, a literature study where performed. The study mostly consisted of reading scientific articles, but also included exploration of similar products and environments.

3.2 Hardware support in developing the prototype

To solve the task at hand, the students together with Capgemini are going to collaborate with Digitalt Laborativt Centrum (DLC) [50] at Halmstad University. DLC have the hardware needed for the project in form of a powerful computer and the VR-headset HTC Vive. In exchange DLC wants to learn from the experiences along the project. Furthermore they want to be able to display the final result at their center.

3.3 The development process regarding the prototype

The project is going to be developed following the iterative prototyping development method. The method consists of an initial identification of user requirements, followed by developing a prototype. Finally an endless loop consisting of using and evaluating the prototype, thereafter revising the prototype [51]. Unity 3Ds own Collab Version Control System (VCS) will be used for software development, to help maintain fully updated versions of the system for all developers, to safely develop and test new functionality for the system without compromising existing ones and to improve documentation of the project.

3.4 Dataset

The students have decided to use a dataset containing Electricity, water, and natural gas consumption [52] of a residential house in Canada from 2012 to 2014 for testing the system. This dataset has been chosen, firstly since it contain a lot of data and also a lot of parameters, giving the students many possibilities of displaying the same dataset. Secondly, dataset contains electricity, water, and natural gas measurements at one minute intervals, there might be some correlation between the weather and energy consumption. The dataset has downloaded and locally stored on the used computer.

3.5 VR development platform

For software development Unity 3D [53] is going to be used, since they are one of the leading VR-engines with good compatibility for both software and hardware, have a big community for troubleshooting and are available for free. Moreover Donalek et al. that's leading in the field of Immersive and Collaborative DV in VR and Paul Sullivan [8], a Master student in Computer Science that wrote his master thesis about graph-based DV in VR: A comparison of user experiences. Both agree on using a supportive game engine

like Unity 3D than building support for HMDs and other components from scratch using OpenGL or similar program. Furthermore Donalek et al. says that their Unity 3D based software could easily and rapidly visualize 10^5 to 10^6 data points, which is comparable to other state of the art visualization tools.

3.6 Data visualization

After exploring the different types of visualizations during the literature study, the students have chosen to use a scatter plot for visualization in their system. Since it's a graph well suited for 3D visualization and displaying many data points. It's also used by many of the leading companies in the field, and hopefully there will be no confusion on understanding the graph like Paul Sullivan [8] had in his thesis. A possible alternative to the scatter plot could have been a node tree. But the students felt that the scatter plot is more capable of displaying many data readings. Another alternative could have been a 3D histogram, but as explained in the theoretical background section, these can become subjects of occlusion.

3.6.1 Data processing

First the local dataset is gathered using the CSVReader script, where it's returned as a Dictionary of strings and objects. Secondly one object of each attribute is tested to see if it's numeric, DateTime or string, since strings have no value and thereby is hard to visualize and compare. These attributes are then shown in the menu system, explained in subsection 3.8.1, for the user to chose between. For the chosen X, Y, Z1, Z2 and Z3 attributes, a minimum and a maximum value is found. Lastly with help of the minimum and maximum values each value is normalized and put in an matrix that later on can be used for the scatter plot. The X and Y values are normalized within their own max and min, while the Z1, Z2 and Z3 values are normalized in regards to their gathered max and min.

3.6.2 Graph rendering

The scatter plot in the prototype consists of prefabricated game objects that are built from Unitys standard shapes and modified by the students. A cube with the scale 0.1 across the different axis and tilted with 45° in the X- and Z-axis, represents a data point in the graph. The cube is tilted to prevent several cubes that have similar position to interfere with each other visually. These prefabricated objects are what the scatter plot is visually. Each object contains a component called Transform, which has information about the objects position, rotation and scale. The X-, Y- and Z-position of the data points are gathered from the normalized matrix. Since the matrix is normalized, the data points position is already relative to each other. All data points share the same X and Y column, and for each additional Z attribute added by the user and ranging from Z1 to Z3, the corresponding column is gather from the matrix. All data points associated to the same Z attribute share the same color to easier separate the attributes visually. To facilitate modification of the data points they're all placed within another game object. Thereby they can all be moved or scaled, just by adjusting their parents transform.

3.7 Functions within the prototype

To fully explore the graph in VR the students believe that a *Interactive* (section 2.3.2) VR environment would suit the prototype. This way the user can move around freely in the environment and also interact with the graph. For this smartphone driven VR would be sufficient. But then movement in the environment would have been limited to 3 DoF and moving in the environment by moving in the real world or looking at objects from above or below would not have been possible. Since the later adds further immersion, the students believe it's vital to fully explore the graph.

External controls are essential to maximize the interaction, with actions such as grabbing, scaling and teleporting. Moreover they are also visible inside the environment, which was not the case of Pauls Sulivans PC-keyboard. This could have led to the slower result of the VR users. Because of these arguments, the students chose to go with a PC driven VR-environment, which also provides a lot more processing power. Since neither they or Capgemini had a system that could be used, their collaboration with DLC became vital. DLC had a HTC Vive and a PC that the students could borrow during the project.

In order to get an interactive prototype, VRTK was used. VRTK is a asset from Unity 3Ds asset store. It is a productive VR Toolkit for rapidly building VR solutions in Unity3d. Several of VRTK scripts are going to be used for basic interaction, like grabbing and scaling the graph, interacting with menus and moving in the virtual world in form of an teleport function. To scale and move the graph, the scripts VRTK.InteractableObject, VRTK.FixedJointGrabAttach and VRTK.AxisScaleGrabAction are beeing used.

3.8 Prototype GUI

In this section the menu system and the components regarding the graph are described.

3.8.1 Menu system inside prototype

To make the graph flexible with different data, a menu system is going to be provided. This way the user can chose what data to been shown on the X, Y, Z1, Z2 and Z3 axis from the dataset. The amount of data points to be drawn will also be flexible. When generating the graph the menu system becomes hidden, not to cause any confusion. Furthermore the user can destroy the graph, making the menu system appear again. Making it possible for the user to draw graphs with different attributes without having to restart the environment.

The menu system is constructed with Unity 3Ds drop-down menu, button and slider and the values is populated and saved within a script. To make the menu interactive with the controllers the VRTK script, VRTK_UI Canvas, is used.

The menu is always at eye-level and facing the user, this is done with some math. First calculating the direction the menu shall face, by taking the coordinates of the menu system and subtracting the coordinates from the headset. With the help of Unitys function, Quaternion.LookRotation(), the rotation of the menu system is set. Furthermore the menu system y-coordinate is set as the same as the headsets y-coordinate, in order to get the menu at eye-level.

3.8.2 Additional information inside prototype

To provide the users with better understanding of the graph suitable axis and information are going to be provided, to further eliminate confusion of the graph. To help guide unfamiliar users, simple tooltips are going to be provided on the handheld controllers. To accomplish this there are three axis showing the X-, Y- and Z-plane. A maximum and minimum point on each axis in text form that displays the appropriate values. A transparent grey cube that encloses the entire plot is also created, this is more an graphical improvement to get an understanding where the chart and data points are located in the room. Along each axis there are a text that displays what's visualized on that axis. Since there are 3 different attributes for the Z-axis, a legend with all visualized attributes is placed above the cube. This keeps the graph from being congested with text. To further help the user understand the new environment, tooltips are shown if the controllers are in the users focus view.

These tooltips are implemented with VRTK_Controller Tooltips script from VRTK. The gray transparent cube, the three axis, the information along the axis and the legend is created within the script, Graph Components. These components is also placed inside the same parent as the data points, so the objects can scale and move together with the data points.

3.9 Basic optimization

When displaying around 6000 data points, there are a lot of information being processed. To prevent the environment from dropping down below 90 frames per second and not to induce any cybersickness, all of the data points are deactivated when moving or scaling the graph, only the transparent cube and the X-,Y, and Z-axis are being displayed, as seen in figure 27. However, to not loose orientation of the data and the chart, some data points can be left in. This will not cause any significant performance reduction.

Cubes vs Spheres - Spheres in Unity 3D contains around 768 triangles, compare to a cube that only contains 12 triangles. By choosing a cube as our data point we decrease the amount of triangles the Graphics Processing Unit have to render dramatically. To further reduce the amount of triangles in the scene, Tetrahedron have been tested. Since they only contain 4 triangles instead of 12. Unfortunately they didn't give the desired result.

Draw calls - A draw call contains all the information telling GPU about textures, states, shaders, rendering objects, buffers, etc. Converting state vectors (all the information mentioned before) to hardware commands for the GPU is very "expensive" for the CPU. Since a draw call is required for each different material, changing and having multiple materials will increase the number of draw calls accordingly. Since CPU work to translate this information to GPU hardware commands takes time, sometimes we see the CPU being a bottleneck to the GPU with a high number of draw calls.

By reducing the number of materials in the scene and to instantiate all datapoints with the same material consecutive, the number of draw calls made by the CPU can be reduced and there by decrease the load, this will increase the prototypes FPS. The information about FPS and draw calls is gathered from Unity's build in Profiler and Stats menu.

3.10 Hardware and Software setup

This subsection covers the hardware and software specifications for the project.

3.10.1 Hardware setup

The workstation used for developing and displaying the prototype is a PC desktop computer running Microsoft Windows 10 Enterprise 64bit operating system and equipped with an Intel Core i7-7800X CPU clocked at 3.50GHz with 6 cores and 12 Logical Processors available for the OS, 16(2x8)GB DDR4 RAM and 256BG SSD + 2TB hard drive. The workstation is also equipped with a NVIDIA GeForce GTX 1080 with GDDR5X.

HTC Vive is equipped with 2 AMOLED displays at 3.6" with an resolution of 1080x1200 pixels per display. Furthermore it has an refresh rate of 90Hz and a FOV of 110 degrees.

3.10.2 Software platform

The developed prototype is based on the Unity 3D game engine. The version used was 2017.2.0f3 Personal, together with Steam VR plugin v1.2.3 and VRTK v3.2.1.

3.11 System evaluation

The prototype system will be analyzed by the students. To see if it fulfills the requirements of visualizing a dataset with VR in 90 FPS, with needed interaction and guiding. The analyze will consist of different parts, testing the environment by the student to see if any visible or physical drawbacks like lag, misorientation of object or cybersickness are shown. To further make sure that maximum amount of data points are shown with acceptable frame rate of 90 FPS, different amount of data points are going to be visualized in the environment using the equipment in subsubsection 3.10.1. To analyze the result of these amounts Unity 3Ds built-in statistics and profiler are going to be used. The result will be shown in a graph in the result. Ideally, an user study would be preformed, where data is presented in a 2D environment and the same data in the developed prototype for two different groups. The groups would then make data readings in their respective environments, then an analyze of the results and presenting them in the results chapter. This user study was not possible due to time constraints.

4 Results

In the result the fundamental parts of the prototype is shown and described. First the handling of the dataset are shown, beginning with all attributes in the dataset, followed by only the numeric ones. Finally the maximum and minimum values used for the normalization and the result of the normalization itself are shown.

Then the menu system inside the prototype are shown. This is followed by how the data points and additional information are drawn. Then an overview of the interaction in the environment are shown. Finally the end product are shown followed by a line plot, showing FPS in relation to the number of cubes in the environment.

4.1 Data processing

Figure 20 shows the top values from the dataset loaded from the local CSV file. As you can see some columns doesn't have any value and some have text in them.

Date/Time	Year	Month	Day	Time	Data Quality	Temp (C)	Temp Flag	Dew Point Temp (C)	Dew Point Temp Flag	Rel Hum (%)	Rel Hum Flag	Wind Dir (10s deg)	Wind Dir Flag	Wind Spd (km/h)	Wind Spd Flag	Visibility (km)	Visibility Flag	Stn Press (kPa)	Stn Press Flag	Hmdx	Hmdx Flag	Wind Chill	Wind Chill Flag	Weather
2012-04-01 00:00	2012	4	1	00:00		4.3		2.3		87		30		28		12.9		99.88						Rain
2012-04-01 01:00	2012	4	1	01:00		4.4		2.2		86		30		17		8		99.94						Rain - Fog
2012-04-01 02:00	2012	4	1	02:00		3.8		1.6		86		30		26		6.4		100						Rain - Fog
2012-04-01 03:00	2012	4	1	03:00		2.8		0.7		86		29		39		8		100.1						Rain - Fog
2012-04-01 04:00	2012	4	1	04:00		2.4		0.4		87		29		30		9.7		100.23						Moderate Rain
2012-04-01 05:00	2012	4	1	05:00		3		0.9		86		29		35		9.7		100.37						Rain
2012-04-01 06:00	2012	4	1	06:00		3.7		1.7		87		27		30		12.9		100.55						Rain
2012-04-01 07:00	2012	4	1	07:00		4.7		1.3		79		26		24		16.1		100.74						Mostly Cloudy

Figure 20: Original values from CSV.

When numeric and date values has been sorted out, this is what the dataset looks like, see figure 21.

Date/Time	Year	Month	Day	Time	Temp (C)	Dew Point Temp (C)	Rel Hum (%)	Wind Dir (10s deg)	Wind Spd (km/h)	Visibility (km)	Stn Press (kPa)
04/01/2012 00:00:00	2012	4	1	00:00:00	4.3	2.3	87	30	28	12.9	99.88
04/01/2012 01:00:00	2012	4	1	01:00:00	4.4	2.2	86	30	17	8	99.94
04/01/2012 02:00:00	2012	4	1	02:00:00	3.8	1.6	86	30	26	6.4	100
04/01/2012 03:00:00	2012	4	1	03:00:00	2.8	0.7	86	29	39	8	100.1
04/01/2012 04:00:00	2012	4	1	04:00:00	2.4	0.4	87	29	30	9.7	100.23
04/01/2012 05:00:00	2012	4	1	05:00:00	3	0.9	86	29	35	9.7	100.37
04/01/2012 06:00:00	2012	4	1	06:00:00	3.7	1.7	87	27	30	12.9	100.55

Figure 21: Non-string values.

To be able to do the normalization of values, minimum and maximum values are found in each column, here in figure 22 are the result of those values.

-	Date/Time	Year	Month	Day	Time	Temp (C)	Dew Point Temp (C)	Rel Hum (%)	Wind Dir (10s deg)	Wind Spd (km/h)	Visibility (km)	Stn Press (kPa)
Min	04/01/2012 00:00:00	2012	4	1	0	1.7	-2.3	46	1	0	6.4	99.88
Max	04/21/2012 19:00:00	2012	4	21	23	16	8.2	90	36	41	48.3	102.65

Figure 22: Min and Max values.

The first rows of the normalized matrix that is used throughout the prototype for plotting and so on, looks like this, see figure 23.

Date/Time	Year	Month	Day	Time	Temp (C)	Dew Point Temp (C)	Rel Hum (%)	Wind Dir (10s deg)	Wind Spd (km/h)	Visibility (km)	Stn Press (kPa)
0	0	0	0	0	0.1818182	0.4380952	0.9318182	0.8285714	0.6829268	0.1551313	0
0.003816794	0	0	0	0.04347826	0.1888112	0.4285714	0.9090909	0.8285714	0.4146341	0.03818616	0.02166249
0.003816794	0	0	0	0.08695652	0.1468531	0.3714286	0.9090909	0.8285714	0.6341463	0	0.04332222
0.007633588	0	0	0	0.1304348	0.07692307	0.2857143	0.9090909	0.8	0.9512195	0.03818616	0.0794227
0.007633588	0	0	0	0.173913	0.04895105	0.2571429	0.9318182	0.8	0.7317073	0.07875895	0.1263558
0.01145038	0	0	0	0.2173913	0.09090909	0.3047619	0.9090909	0.8	0.8536586	0.07875895	0.176897
0.01145038	0	0	0	0.2608696	0.1398601	0.3809524	0.9318182	0.7428572	0.7317073	0.1551313	0.241879
0.01526718	0	0	0	0.3043478	0.2097902	0.3428572	0.75	0.7142857	0.5853658	0.2315036	0.3104691
0.01526718	0	0	0	0.3478261	0.2517483	0.3428572	0.6590909	0.5714286	0.4878049	0.4224344	0.3826728
0.01908397	0	0	0	0.3913043	0.2937063	0.4	0.6590909	0.4571429	0.3170732	0.4224344	0.4296031
0.01908397	0	0	0	0.4347826	0.3706294	0.3619048	0.5	0.6	0.1707317	0.6157518	0.4584846
0.02290076	0	0	0	0.4782609	0.4265734	0.2857143	0.3409091	0.6285715	0.1463415	0.6157518	0.505415
0.02290076	0	0	0	0.5217391	0.5174825	0.4	0.3409091	0.6285715	0.09756097	0.6157518	0.5306855
0.02671756	0	0	0	0.5652174	0.5454546	0.247619	0.1590909	0.6285715	0.2195122	0.6157518	0.552348
0.02671756	0	0	0	0.6086956	0.5664335	0.0952381	0	0.9142857	0.09756097	0.6157518	0.5776187
0.03053435	0	0	0	0.6521739	0.6153846	0.2380952	0.06818182	0.6571429	0.1707317	0.6157518	0.6064973
0.03053435	0	0	0	0.6956522	0.5454546	0.4190476	0.3181818	0.6	0.4634146	0.6157518	0.6173272
0.03435114	0	0	0	0.7391304	0.4895105	0.4476191	0.4318182	0.5142857	0.6341463	0.6157518	0.6534305

Figure 23: Normalized values in Matrix.

4.2 Prototype system for DV with VR

In this section the the different features, the graph, some interactions and the final results are shown.

4.2.1 Feature selection

In the menu the attributes for each axis can be selected, as well as the number of data points being shown in the generated graph. This is shown in figure 24.



Figure 24: The initial menu system.

4.2.2 Graph visualization

When the X, Y, Z1, Z2 and Z3 attributes has been chosen, the graph is drawn, see figure 25.

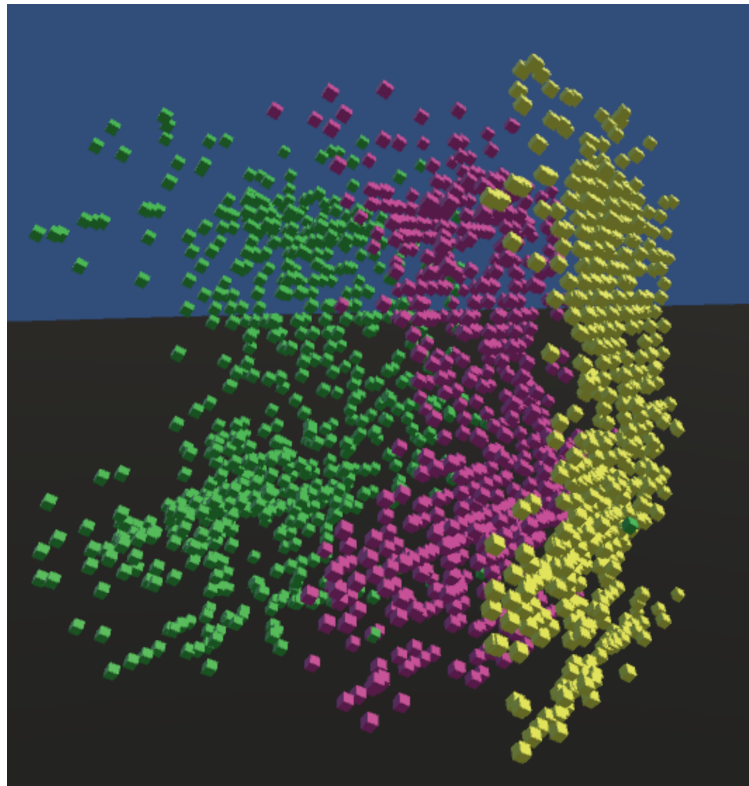
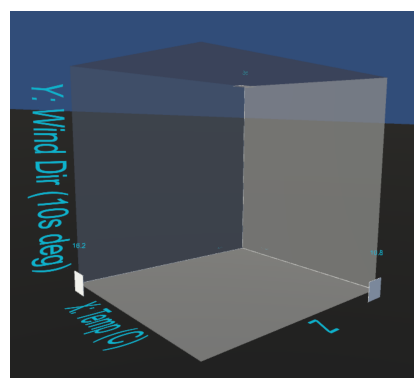


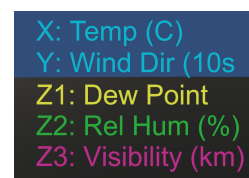
Figure 25: Scatter plot without axis and frame.

4.2.3 Additional information inside prototype

Below in figure 26 are the graphical improvements to help the user visualize and understand the graph is shown. Figure 26(a) shows the box surrounding the graph. Figure 26(b) shows the legend with axis information that's placed above the graph



(a) The scatter plot frame.



(b) Legend for graph attributes.

Figure 26: In (a) the framework for the graph is shown. In (b) the legend is presented, it is a concise explanation of the symbols used.

4.2.4 Interaction tools

Below, a scaled version of the cube is shown in figure 27, scaling can be done using grabbing with both controls and then dragging them apart. Furthermore the tooltips of the controller is shown in figure 28, this happens when it's in the users focus view. Teleport is used to move around in the environment without moving physically. UI selector is a laser-pointer which is used when selecting axes.

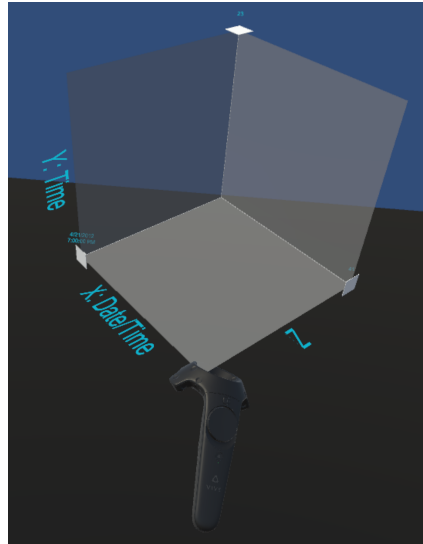


Figure 27: A scaled graph frame, grabbed with one of the controllers.

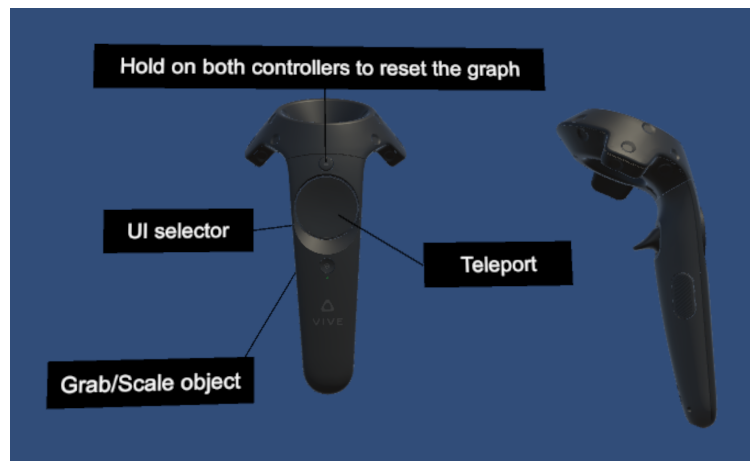


Figure 28: The controllers which is showing its tooltips since it's in the users focus view.

4.2.5 Final results

Below in figure 29 is a scatter plot in the finished prototype is shown. This specific plot shows X: Temp (C), Y: Wind Dir, Z1: Dew Point (Yellow), Z2: Rel Hum(%) (Green) and Z3: Visibility (km) (Purple) using 500 readings from each attribute.

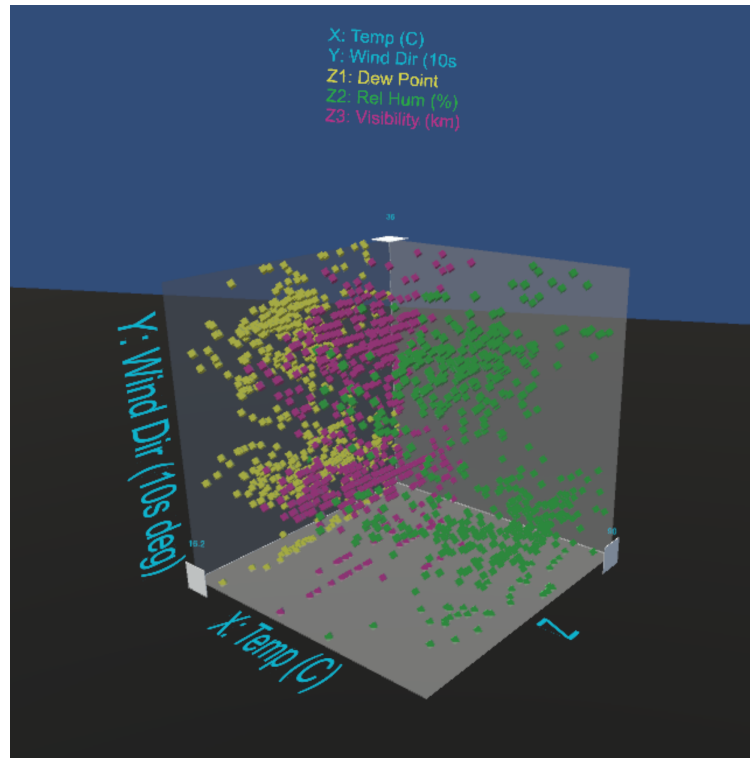


Figure 29: Prototype of scatterplot with X: Temp (C), Y: Wind Dir, Z1: Dew Point (Yellow), Z2: Rel Hum(%) (Green) and Z3: Visibility (km) (Purple) spanning over 500 reading for each attribute.

4.3 Analyze of results

Figure 30 shows the statistics of FPS in relation to the number of cubes drawn in the environment, where the red line indicates 90 FPS. Note that this is the reciprocal FPS. Thereby it only includes the time taken to do the frame update and render the game view. It does not include the time taken in the editor to draw the scene view, inspector and other editor-only processing. The data are gathered from Unity 3Ds statistics.

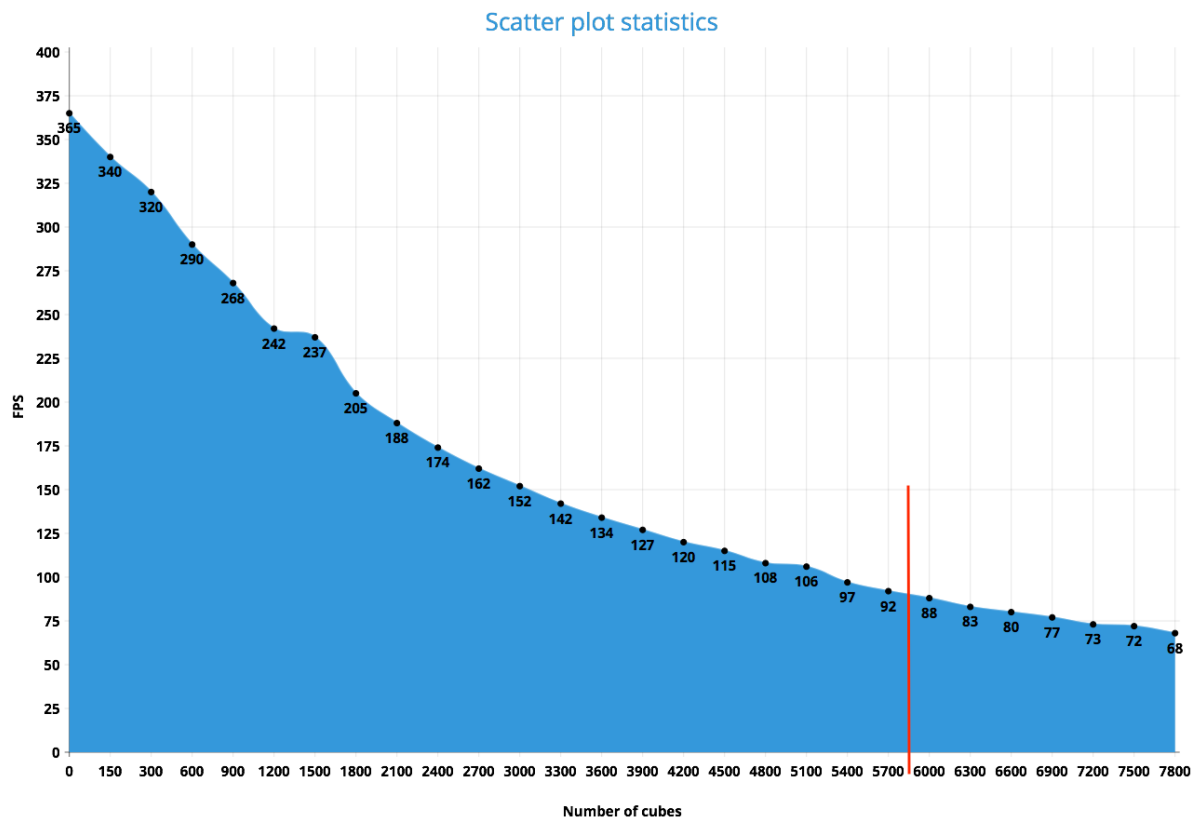


Figure 30: Data gathered from Unity 3Ds statistics, showing FPS in relation to the number of cubes that are drawn in the environment.

5 Discussion

In this chapter the project are evaluated and discussed by the students.

5.1 Theory

Since the requirements of the project where very vague, we didn't exactly know what to research and at what depth. Therefore the theory part is very wide and covers many aspects of DV and VR. We still think that we've covered the concerned parts about DV, VR, platforms and interaction in regarding the first purpose, to gain knowledge about existing techniques, frameworks, hardware and software.

To fully extend a VR experience, we believe that 6 DoF are needed. This way the experience get's more real and it could also prevent cybersickness in some manners. Otherwise moving in the real world, doesn't affect anything in the virtual world. Which could cause motion sickness, disorientation and so on. Regarding HMDs it depends on the product. A smartphone driven HMD could be enough for a simpler application or game, even though they only provide 3 DoF. For maximal processing power, a PC driven HMD should be used. The HTC Vive is easy to develop for since there's a lot of support for it. Although it would be nice to see how different the WMR HMDs are, and if the standalone HMDs stand a chance to the PC driven ones. We feel that the outside-in tracking of the HTC Vive is really accurate, but we have not tried or experienced any other tracking methods.

We recommend, just as the reports in related work, to use an game engine like Unity 3D. Since this rapidly speeds up the development process. For displaying 2D objects in VR, we recommend to always have these facing the user if possible. Even though VR is three-dimensional, it may not always be beneficial to visualize 3D objects. Since distortion or occlusion can occur when for example choosing a 3D pie chart or a 3D histogram. We also believe that an informative and non complex interaction in the environment is needed, since few people are used to VR itself, and even less to the specific product. Moreover it's really important to minimize the chances of cybersickness, since this can impair the whole experience. VRs greatest strengths lies in immersion and interaction, to make the virtual experience as good as possible, one must take advantage of these.

5.2 Prototype

We believe that the prototype is using VRs advantages over traditional DV when it comes to immersion and interaction, which could give the user advantages in data analysis. If the system instead would have been using a smartphone driven VR-environment, we don't think that the advantages to traditional DV would have been as clear. Since it would only allow for 3 DoF.

Even if 6 DoF is available, it may not always be possible for the user to setup a big enough play area. Which has a minimum of 2x1,5m for the HTC Vive. Otherwise only a seated/standing environment is playable. In these cases or when the user wants to walk great distances inside the environment, the teleport function is of great essence. Moreover we think that the tooltips is enough informative and isn't disturbing the exploration of the graph.

We think that a scatter plot as visualization form/technique was the right choice. The graph is able to show a lot of readings and are widely known in data analysis. Even though it might be harder for less experienced users to understand the data, then compared to a histogram. The idea was to emphasize how we can actually interact with data and to show how multidimensional we can view data. Not only can you use the three axis (X, Y, Z) you can also use shape and size to differentiate data points.

One drawback with a scatter plot is that each data point need to be represented in 3D. In a scatter plot there can be a large amount of data points, to see trends and patterns in the data. Which makes it almost impossible to draw spheres since they consist of so many triangles. Cubes are acceptable, but still not ideal. We've experimented with Tetrahedron and other shapes to reduce the triangles needed to be drawn, to be able to keep a good refresh rate at 90 fps. But haven't got it working properly. Even though few materials is used and all datapoints with the same material are instantiated consecutive to save a lot of draw calls, we still can't visualize over 10 000 data points using cubes. This could have been researched/tested earlier in the deployment to maybe chose another way of visualization. But since Donalek et al. says that their Unity 3D based software could easily and rapidly visualize 10^5 to 10^6 data points, we didn't think this was going to be a problem. This also supports the choice of PC driven VR, since a lot more computational power is available.

5.3 Project

The prototype system has been developed with the prototyping development method, which feels like an appropriate method, since it is in fact a prototype that's been developed. Meaning no previous knowledge about this specific product exist, neither does any clear demands or requirements.

Testing and evaluation during the project, has mainly consisted of interaction with the prototype by the students and with Unity 3Ds built in Profiler statistics. Since few functional and non-functional requirements were provided, the testing has been limited. If for example, the aim of the project would have been to optimize the rendering time or to compare different types of hardware or software, testing could have been more comprehensive.

We believe that the project is having a academic level. A lot of research has been done in the field both during the pilot study and during development. Since Unity 3D is handling a lot of things like rendering, graphics, physics and so on by itself, we haven't looked into these things explicitly. Therefore they are not explained explicitly in the report either. Even if they would have held a large mathematical depth.

5.4 Social requirements for technical product development

Economically and environmentally our prototype may not be so beneficial, since the equipment to run it is quite expensive and not very environmental friendly. Although, normal data analysis program also need to be run with an expensive and non environmental friendly equipment. Hopefully our prototype can help data analysts to easier spot trend and patterns compared to traditional DV and can thereby be economical helpful and possibly even environmental friendly in other aspects. We're not crossing any security or privacy concerns, since it's up to the user to choose what data to visualize. The user must also already have access to the data in terms of a CSV-file.

6 Conclusion

In regard of the first purpose, to gain knowledge about different technologies of DV through VR, we have in subsection 1.4 explained what Data Visualization and Virtual Reality is and showed similar solution. In section 2 we investigate more in depth about different visualization techniques and technologies and frameworks for VR. The second purpose of the project is to develop a prototype system for Data Visualization in Virtual Reality. The methodology for this is described in section 3 and the result is presented in section 4.

Although the prototype can be improved in many aspects. It uses the 6 DoF to its benefit by allowing the user to move inside the environment by moving in the real world. It also provides additional interaction with the graph using the controls, which makes it a lot more interactive and flexible. The technology itself provides a very immersive environment compared to traditional displays.

We consider that we have answered all the questions in subsection 1.2 in a good and understanding way through out the report.

By utilizing Unity 3D as our development platform for our prototype, we could easily begin testing, planning and experimenting with different solutions. Using a game engine for this task was a good choice, compared to building a similar program for basic functionality using OpenGL.

In regarding Data Visualization through Virtual Reality we can see the presumed benefits by using VR. Unfortunately we cannot see this well enough in our prototype to show any clear result. For this an user study would have been needed, to compare the same data in VR and with traditional DV. To examine accurateness and quickness of the users.

6.1 Future optimization

There are a few things that could be added to make the prototype better in terms of interaction and understanding. Some way of showing the value of a specific data point chosen by the user. Since the data from the dataset is processed, the system could easily be extended to show different kinds of graphs. Furthermore it would have been nice to be able to chose a dataset from within the environment and not be limited to the one chosen from the Unity 3D editor. Moreover we believe that Machine Learning could have been implemented with the system to help with the user to see trends and patterns.

Bibliography

- [1] Capgemini website. <https://www.capgemini.com>. Accessed: 2018-01-29.
- [2] Mathematica plot. <http://reference.wolfram.com/language/ref/Plot.html>. Accessed: 2018-01-26.
- [3] Matlab plot. <https://se.mathworks.com/help/matlab/ref/plot.html>. Accessed: 2018-01-26.
- [4] Tableau. www.tableau.com. Accessed: 2018-03-08.
- [5] Qlik. www.qlik.com. Accessed: 2018-03-08.
- [6] Ciro Donalek, S. George Djorgovski, Scott Davidoff, Alex Cioc, Anwell Wang, Giuseppe Longo, Jeffrey S. Norris, Jerry Zhang, Elizabeth Lawler, Stacy Yeh, Ashish Mahabal, Matthew J. Graham, and Andrew J. Drake. Immersive and collaborative data visualization using virtual reality platforms. *CoRR*, abs/1410.7670, 2014.
- [7] Virtualistics website. <https://www.virtualitics.com>. Accessed: 2018-01-31.
- [8] Paul Sullivan. Graph-based data visualization in virtual reality: A comparison of user experience. *the Faculty of California Polytechnic State University*, 2016.
- [9] Virtual reality is here. the possibilities are endless. <https://nirvaniq.com/>. Accessed: 2018-05-01.
- [10] Lookvr is an experiment in exploring data in new ways. <https://looker.com/platform/blocks/embedded/lookvr>. Accessed: 2018-05-01.
- [11] Rikke Friis Dam Mads Soegaard. *The Encyclopedia of Human-Computer Interaction, 2nd Ed.*, chapter 35. Data Visualization for Human Perception by Stephen Few. The Interaction Design Foundation, 2013.
- [12] S. Dübel, M. Röhlig, H. Schumann, and M. Trapp. 2d and 3d presentation of spatial data: A systematic review. pages 11–18, Nov 2014.
- [13] Michael Friendly. Visions and re-visions of charles joseph minard. *Journal of Educational and Behavioral Statistics*, 27(1):31–51, 2002.
- [14] Points in the coordinate plane. <https://www.ck12.org/algebra/Points-in-the-Coordinate-Plane/lesson/Points-in-the-Coordinate-Plane-ALG-I/>. Accessed: 2018-05-07.
- [15] A graphical representation of the wind on a map. <http://hint.fm/projects/wind/>. Accessed: 2018-05-07.
- [16] Map projections that lie. <http://geoffboeing.com/2015/08/map-projections-that-lie/>. Accessed: 2018-05-07.
- [17] Charles minard’s map of napoleon’s disastrous russian campaign of 1812. <https://www.edwardtufte.com/tufte/minard>. Accessed: 2018-05-07.

- [18] Visually blog. <https://visual.ly/blog/2ds-company-3ds-a-crowd/>. Accessed: 2018-03-5.
- [19] Matlab plot - scatter plot 3d. <https://www.mathworks.com/matlabcentral/fileexchange/35288-matlab-plot-gallery-scatter-plot-3d?focused=6793187&tab=example>. Accessed: 2018-05-07.
- [20] Visualizing a complex function. https://math.stackexchange.com/questions/463174/how-to-slice-re1-lz-into-a-cartesian-function-for-any-angle?noredirect=1#comment2261223_463174. Accessed: 2018-05-07.
- [21] Finite element method (fem). <https://cadsetterout.com/inventor-tutorials/get-started-with-autodesk-inventor-stress-analysis/>. Accessed: 2018-05-07.
- [22] David Zeltzer. Autonomy, interaction, and presence. *Presence: Teleoperators and Virtual Environments*, 1(1):127–132, 1992.
- [23] Tomasz Mazuryk and Michael Gervautz. Virtual reality history, applications, technology and future. 1999.
- [24] G Franck C Ware. Viewing a graph in a virtual reality display is three times as good as a 2d diagram. *Visual Languages, 1994. Proceedings., IEEE Symposium*, 1994.
- [25] H. Scott Fogler. The investigation and application of virtual reality as an educational tool. *Proceedings of the American Society for Engineering Education 1995 Annual Conference*, 1995.
- [26] M. abrash, “what vr could, should, and almost certainly will be within two years,” january 2014. <http://media.steampowered.com/apps/steamdevdays/slides/vrshouldbe.pdf>. Accessed: 2018-02-26.
- [27] S. Hahn T Proffen J Daniel M Matheson M. Drouhard, C. A. Steed. Immersive visualization for materials science data analysis using the oculus rift. *2015 IEEE International Conference on Big Data (Big Data)*, 2015.
- [28] G J Kim A Rizzo. A swot analysis of the field of virtual reality rehabilitation and therapy. *Presence: Teleoperators & Virtual Environments*, 14(2):119–146, 2005.
- [29] Vr excavate blog. <http://vrexavate.blogspot.se>. Accessed: 2018-03-06.
- [30] Z Zhou X Ma R Briggs J Baileym G Kennedy S O’Leary S Wijewickrema, B Copson. Design and evaluation of a virtual reality simulation module for training advanced temporal bone surgery. *Computer-Based Medical Systems (CBMS), 2017 IEEE 30th International Symposium*, 2017.
- [31] S. Asada R. Burkhard B. Klein S. Mueller Arisona S. Schubiger J. Perhac, W. Zeng. Urban fusion: Visualizing urban data fused with social feeds via a game engine. *Information Visualisation (IV), 2017 21st International Conference*, 2017.
- [32] Jonatan Berglund Liang Gong Björn Johansson Johan Vallhagen Daniel Nåfors, Erik Lindskog. Realistic virtual models for factory layout planning. *Simulation Conference (WSC), 2017 Winter*.

- [33] I Polasek J Vincur, P Navrat. Vr city: Software analysis in virtual reality environment. *Software Quality, Reliability and Security Companion (QRS-C), 2017 IEEE International Conference*, 2017.
- [34] R A. Calvo Y Wang, K Ijaz. A software application framework for developing immersive virtual reality experiences in health domain. *Life Sciences Conference (LSC), 2017 IEEE*, 2017.
- [35] Yun-Xuan Lin Wen-Chieh Lin Kwan-Liu Ma Yi-Jheng Huang, Takanori Fujiwara. A gesture system for graph visualization in virtual reality environments. *Pacific Visualization Symposium (PacificVis), IEEE*, 2017.
- [36] Google cardboard. <https://vr.google.com/cardboard/>. Accessed: 2018-05-07.
- [37] Google daydream. <https://vr.google.com/daydream/>. Accessed: 2018-05-07.
- [38] Samsung gear vr. <http://www.samsung.com/global/galaxy/gear-vr/>. Accessed: 2018-05-07.
- [39] Positional tracking technology. http://www.latticesemi.com/en/Blog/2018/03/12/16/31/VR_Tracking. Accessed: 2018-05-02.
- [40] Hiren bhinde (qualcomm): On-device motion tracking for immersive vr. <https://www.slideshare.net/AugmentedWorldExpo/hiren-bhinde-qualcomm-ondevice-motion-tracking-for-immersive-vr>. Accessed: 2018-05-02.
- [41] Htc vive. <https://www.vive.com/eu/>. Accessed: 2018-05-07.
- [42] Hp windows mixed reality headset. <https://store.hp.com/us/en/cv/mixed-reality-headset>. Accessed: 2018-05-07.
- [43] Hmd odyssey - windows mixed reality headset. <https://www.samsung.com/us/computing/hmd/windows-mixed-reality/xe800zaa-hcl1us-xe800zaa-hcl1us/>. Accessed: 2018-05-07.
- [44] Oculus go. <https://www.roadtovr.com/gdc-2018-oculus-go-to-support-fixed-foveation-and-up-to-72hz-refresh/>. Accessed: 2018-04-28.
- [45] Lenovo mirage solo. <https://www3.lenovo.com/us/en/virtual-reality-and-smart-devices/virtual-and-augmented-reality/lenovo-mirage-solo/Mirage-Solo/p/ZZIRZRHVR01>. Accessed: 2018-04-28.
- [46] 6 degrees of freedom. <http://www.leadingones.com/articles/intro-to-vr-4.html>. Accessed: 2018-04-28.
- [47] Unity 3d platforms supported. <https://unity3d.com/unity/features/multiplatform>. Accessed: 2018-05-01.
- [48] The deprecation process for unityscript. <https://blogs.unity3d.com/2017/08/11/unityscripts-long-ride-off-into-the-sunset/>. Accessed: 2018-05-01.

- [49] Unreal engine plattform support. <https://www.unrealengine.com/en-US/faq>. Accessed: 2018-05-01.
- [50] Digitalt laborativt centrum (dlc) website. <http://www.hh.se/akademinforlarandehumanioraochsamhalle/digitaltlaborativtcentrumdlc.65445572.html>. Accessed: 2018-01-29.
- [51] Maryam Alavi. An assessment of the prototyping approach to information systems development. *Communications of the ACM*, 27(6):556–563, 1984.
- [52] Electricity, water, and natural gas consumption dataset. <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/FIE0S4>. Accessed: 2018-01-26.
- [53] Unity 3d website. <https://unity3d.com>. Accessed: 2018-03-06.

A Time Table

Down below is the time table for the project. Milestone 6, 7 and 11 have been altered as seen in purple.

Project: Virtual Reality Data Visualization										Date: 2018-03-09		Reviewed:													
Client: Capgemini										Version: 1.2															
Course: Theis project										Issued by: Oliver & Mattias															
Activities				Date	Hours	Student		Time plan (weeknumber)																	
Nr	Description					Initials	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1	Project plan						46	10																	
2	Pilotstudie						8	20	50	15	15	10	2												
3	Development									5	6	5	5	5	15	35	10	35	12	12	6				
4	Testing														10	12	12	12	15						
5	Integration														10	13	11	11	14						
6																									
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8																									
9																									
10																									
11																									
12	Seminars & Lectures																								
13	Report																								
14	Supervisormeeting																								
15																									
16	Milestones																								
17	Project plan done																								
18	Project plan seminar																								
19	Decide game-engine and headset																								
20	Lecture writing a report																								
21	Kanban board / detailed timeplan done																								
22	Extract data from dataset to game-engine																								
23	First prototype with visualized data on graph																								
24	Halfway report done 12:00																								
25	Halfway seminar																								
26	Evaluate feedback halfway seminar																								
27	Manipulate objects in VR																								
28	Prototype ready for final tests																								
29	Final product done																								
30	Final report																								
31	Final seminar																								
32	Individual examination																								
33	Last date for edited report																								
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40																									
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							58	47	59	22	28	29	53	30	40	40	40	40	46	40	41	41	10	1	69

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